

IN ONE WORD: NOT FROM EXPERIENCE *

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Studies on clinical inference show that people do not always improve their judgments with experience. This paper argues that the expectation that they will improve is mistaken and founded on an incorrect conception of the nature of experience. Changing this conception towards a more adequate one along the lines suggested by Popper (1963) leads to a far more pessimistic view about people's ability to learn from experience, a view that is in closer correspondence with the facts from studies on clinical judgment. The paper also reviews results from psychological studies concerned with people's ability to learn from experience in probabilistic situations. These studies show that people have a number of biases which prevent them from using the information which experience provides. Examples of such biases are the tendency to use confirmatory evidence, assumptions about causality, and disregard of negative information. The paper argues that these biases can be understood in terms of the kind of information that people actually have to use when learning from experience outside the laboratory.

In 1959, Goldberg published a study in which he compared experienced clinical psychologists and secretaries with respect to their ability to make diagnoses of brain damage. The diagnoses were made from the Bender Gestalt Test, an instrument widely used for making diagnoses of this kind. The results were as clear as they were surprising: there were no differences between the two groups, clinical psychologists were no better than secretaries in making these diagnoses.

Although these results are particularly striking, Goldberg's study is but one of many studies showing that even considerable experience with a clinical task does not make people very good at making the decisions required by this kind of task (see, for example, Goldberg 1968, for a review of this research).

One reaction to this kind of findings has been to consider them con-

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firmation of one's worst suspicions about clinical psychologists and psychiatrists. A more interesting way of looking at them is, however, to remind oneself that even psychiatrists and clinical psychologists are people just like the rest of us, and to note that we have in this literature a solid body of evidence that people do not always learn from experience, at least not when the experience consists of a series of cases. This may be a startling conclusion and certainly one that goes against some of our most cherished beliefs: that experience improves our judgments and decisions.

It is possible to understand the conviction that we learn from experience in terms of its roots in British Empiricist philosophy, but the strength of the conviction suggests that it may serve not only our scientific understanding, but that it has ideological functions as well (see Greeno *et al.* 1978 for a discussion of this point). It is, however, a conviction that at least psychologists have not been able to keep without considerable cost. Part of this cost is that we have come to have what can only be called a perverse conception of the nature of experience.

This conception is very clearly illustrated in the model of experience which is manifest in psychological experiments on learning, and it is nowhere more clearly expressed than in the so called paired-associates paradigm. In experiments in this paradigm, the subject is given a list of word pairs and asked to learn to give the second member of each pair as his response when presented with the first member as a stimulus. This is a task which subjects learn with little difficulty, thus seemingly vindicating the hypothesis that experience leads to valid knowledge. What these results show, however, is only that experience can lead to a modification of a person's mind. It does not show that the knowledge obtained in this manner will be valid, for as an object of knowledge, there is something very special about a list of paired associates. This is that we know exactly what there is to be learned from this list. In the words of Karl Popper (*e.g.*, 1963), the peculiar thing about such a list is that truth is manifest. The process of acquiring knowledge therefore becomes a simple matter of creating a mental picture of the task, and if this picture is a faithful one, it also constitutes knowledge about the world. Thus, in this case, experience does indeed lead to valid knowledge.

But note why this is the case. The guarantee of validity is *not* in experience itself. There is nothing in the subject's experience with the task that even suggests that a given stimulus work will be followed by

the same response word on each and every trial. Clearly, the guarantee of validity does not come from experience. It comes from the *experimenter* who has assured the subjects, not only that there *is* something to learn, but also *what* there is to learn. This, rather than experience as such, guarantees the validity of the knowledge acquired in this paradigm. The paradigm may thus very well model the situation in teaching, where the teacher decides for the pupil what the truth should be in a given case, but it certainly does not model the situation in which a person is learning from experience.

Take the case of a clinical psychologist learning about diagnoses and treatment from his personal experience with patients. In this case, nobody will be able to tell him what there is to learn, or even whether there is anything for him to learn. He may learn what his colleagues think the correct diagnosis and treatment should be, but this is only learning about what another person thinks that he knows, but not about the patients. And just because the kind of treatment selected turns out to lead to recovery does not mean that the decision was correct for this recovery may have many causes other than the treatment given and, of course, many other kinds of treatment might have been just as effective. Take for example, the belief in the efficacy of penicillin as a cure for sore throats! Penicillin, as we know, is a good cure for diseases caused by bacteria, but sore throats are usually caused by viruses rather than bacteria and for these, penicillin is not effective. Nevertheless, the patient recovers, but not because of the penicillin but because of the body's general ability to fight disease.

In these kinds of situations, then, truth is no longer manifest. Indeed, this is only the beginning of our problems. It is not only the case that the truth is not readily apparent; we may also distort the situation by our actions, so that what we could have learned will not be learned. We will return to this problem later in the paper, and we will start with the simpler case when we do not act but learn from just observing the world.

Learning from observing the world

The problem of induction

The psychological problem of learning from experience is, of course, only the old philosophical problem of induction. As philosophers after Hume have agreed, inductive judgment cannot be justified in the way

deductive judgment can be justified. The judgment may very well be true, and serve as a guide to action, but it cannot be *shown* to be true. The fact that it works says little about its truth, it just tells us that it works, and the explanation for why it works may be very different from what we think it is.

Whereas philosophers have given a lot of thought to the problem of induction, psychologists have generally not given much thought to the corresponding problem of learning from experience. They have been content to note that the behavior of people changes as a result of experience, but they have not considered whether the persons also have acquired more valid knowledge; this has been taken for granted. Nor have they considered the problem of where our faith in experience may come from, given that the things we learn cannot be justified. Instead, they have just accepted Hume's psychological theory of induction. This theory holds that as we experience may things of a similar kind, a 'habit of mind' makes us believe that we will see even more things of this kind. Thus, when we have seen five white swans, we tend to believe that the next swan will also be white.

The objections to this theory are well known and have to do with the problem of similarity (Popper 1963). It is all very well to say that our belief is strengthened when we see many similar things, but how do we know that the things that we encounter are really similar and instances of the same phenomenon? This may not be very much of a problem in a traditional psychological experiment on induction, such as a concept learning experiment where the things are colors and simple geometrical figures, stimuli that our evolutionary history has taught us to recognize. But what about the more complex stimuli encountered in the real world such as psychiatric patients, or political decision problems? Here, we have to contribute to the learning process by first deciding what things are similar and what things are not similar to form a basis for our induction. But for that, we need to have an idea of what the relevant characteristics are, *i.e.*, we need to know what characteristics will lead to a useful classification. In short, we need to define what we are to learn, before we are able to learn it.

The learning process, therefore, cannot simply be a case of forming a mental picture of the world. It must be an active process of hypothesis testing. This leads to two problems for research; *viz.*, the problem of where our hypotheses come from, and that of how we test our hypotheses. We now turn to a brief review of what psychological research has to say on these two problems.

Hypothesis testing theories

Hypotheses testing theories have a long history in experimental psychology. Interesting enough, the first theory of this kind was developed for discrimination learning in the white rat (Krechevsky 1932) at a time when most psychologists would not allow a person to think or have hypotheses for fear of being accused of mentalism. This theory then came to set the pattern for subsequent theories of this kind in psychology especially those within the area of concept learning. The most well known is, of course, that of Bruner *et al.* (1956).

The distinguishing characteristic of the situation for which these theories have been developed, is that the subjects' task is to discover which of the dimensions of a series of stimulus objects are useful for dividing them into two or more classes.

For these situations, it is not difficult to decide where the subjects' hypotheses come from. The hypotheses are simply sampled from the stimuli. The theories assume that the subjects are able to analyze the stimuli into their relevant components, and that they then try the usefulness of these components, either one by one, or in some combination. These theories are thus only new versions of the old abstraction theories for concept formation. In their actual application, the theories tend to bypass the problem of abstraction, however, in that they are applied to the learning of stimuli, the components of which are already well formed concepts, such as color and form. These experiments, therefore, are set up in such a way that the experimenter is certain that the subjects already have the hypotheses relevant to the task. This means, however, that these experiments do not tell us what we need to know, and the extension of these theories to situations where the components are not well learned concepts becomes very problematical. In short, we do not know whether these theories really apply to the learning of new concepts (see Bolton 1972).

This is not the place to review all the results obtained with the experimental paradigm that have grown out of this conceptualization of concept learning, especially since there are already good reviews available (see, for example, Bourne 1966, for a short introduction). One result, however, is of great interest in the present context. This is that subjects learning these kinds of tasks do so by trying to confirm hypotheses, rather than by trying to refute incorrect hypotheses. Furthermore, they rely mainly on those tests which provide them with positive

information about the concept, *i.e.*, information telling them what the concept is, but neglect negative information, *i.e.*, information telling them what the concept is not (see Bruner *et al.* 1956, and the paper by Mynatt *et al.* 1977).

Confirmation will, of course, not teach the subjects about the actual validity of their hypotheses; it will only tell which hypotheses work, although the reason why the hypotheses work may be very different from what the subject thinks. In fact, it is clear that this kind of strategy may actually prevent the subjects from discovering even rather simple relations in the task. An experiment by Wason (1960) gives a good illustration of this. In Wason's experiment, subjects had to find a simple rule by generating triplets of numbers. The rule was a simple monotonic series, so that any three numbers in an ascending order was correct, *e.g.*, 1, 2, 3, or 1, 501, 10 000. However, an infinite number of complicated series will fulfill the condition of monotonicity, and subjects would use all sorts of rules, which fulfilled this condition, thus arriving at correct answers. Nevertheless, their rules would, however, not be the correct rule, and a majority of subjects actually failed to find the simple rule of monotonicity because they sought confirmation of a rule they believed to be correct, rather than refutation of a possibly incorrect hypothesis.

In concept learning experiments, the neglect of negative information leads to slow and inefficient learning, but the strategy makes sense when one considers the conditions under which people usually have to learn in the so-called real world. Negative information is clearly useful in a concept learning experiment where the subjects know the dimensions of the 'world'. For example, if there are red and green squares, knowing that a green square is not an instance of the concept immediately tells us what the concept is. But to be told that a cat is not an elephant when one does not even know how many different kinds of animals there are, will not teach us very much about elephants. Being told what an elephant is, on the other hand, is highly useful. But note that this is the case only because there is a guarantee of truth in the very fact of being told. This can be given because this statement informs a person, not about the world as such, but about the kinds of concepts people have.

When we have to learn from outcomes, it may, in fact, be almost impossible to discover that one really does not know anything. This is especially true when the concepts are very complex in the sense that each

instance contains many dimensions. In this case, there are too many ways of explaining why a certain outcome occurred, and to explain away failures of predicting the correct outcome. Because of this, the need to change may not be apparent to us, and we may fail to learn that our rule is invalid, not only for particular cases but for the general case also. This is very clearly demonstrated in a series of experiments by Fischhoff and Slovic (1978). Their experiments show that even very limited amounts of experience with complex stimuli, such as handwriting samples and common stocks, lead to a highly unwarranted confidence in one's ability to classify these stimuli when the subjects believe that they are in possession of a good rule for making the classifications. Indeed, these results suggest that the mere fact that they are able to find a rule is sufficient for the subjects to believe that they can make the appropriate discriminations even though they have no experience indicating that the rule is valid.

However, even though the hypothesis-testing theories mentioned above are a step towards a general theory of how people classify stimuli, and thus also a step towards a theory of how people learn from experience, they are clearly not sufficient. This is because they do not cover the important problem of how relations between variables are learned. We now turn to this problem.

Learning relations between variables

A typical judgment task, such as a diagnostic problem in clinical psychology, requires not only that one learns to classify patients but also, and perhaps to a greater extent, that one learns the relations between variables, *e.g.*, the relations between various symptom configurations and treatment outcomes. The process of learning the relations in tasks of this sort, *e.g.*, the relation between MMPI scores and degree of illness, is made difficult by the fact that the relations among the variables are probabilistic rather than deterministic.

These tasks differ from the classification tasks discussed above in that it is not even theoretically possible to construct hypotheses by sampling aspects from the stimulus configurations. A relation between two variables is not defined by any single instance, and even if multiple instances are available, they have to be ordered before the relation can be detected. Such an ordering does, of course, presuppose a hypothesis about the nature of the relation being looked for, so ordering does not provide any automatic solution.

Since the hypotheses about the nature of the relation between variables cannot come from the stimulus objects, they have to come from the subjects. That is, the subjects have to contribute the hypotheses. It is thus clear that what will be learned in this case will very much depend on what kinds of hypotheses the subjects will bring to the task.

In a series of studies, we have used various forms of probabilistic inference tasks to investigate this problem. The tasks have required the subjects to learn to use one or more cues to infer the state of a criterion variable which is imperfectly correlated with the cues. The only information given to the subject is the correct criterion value on every trial, so learning takes place on the basis of outcome feedback only.

The first few studies (Brehmer, 1974; Brehmer *et al.* 1974) were directed at the problem of how subjects find the functional relations between two variables. The results agreed quite closely with the predictions from a hypothesis sampling model based on the assumption that subjects have a limited number of hypotheses which are ordered in a hierarchy according to strength, and that they sample and test hypotheses in the order determined by the relative strengths of these hypotheses. Later experiments have shown, however, that subjects are also able to construct some new hypotheses not in their hypothesis hierarchy, and that they are thus not limited to learning only the few rules about which they have hypotheses in their hierarchy (Brehmer 1976a, 1980; Brehmer and Kuylensstierna 1979). But even though such constructions are possible, linear hypotheses are by far the dominant ones. This has been interpreted to mean that linear relations are the most common relations in real world inference tasks and subjects have therefore learned to try this rule first (Brehmer 1974). However, this hypothesis would go against the main thesis of this paper. The results of Dawes and Corrigan (1974) suggest a different interpretation more in line with the present argument that people do not learn from experience.

Dawes and Corrigan showed that a linear model, and even one with unit weights, provides a good approximation to most judgment tasks. If equipped with such a model, the person is simply not likely to find any better model. It is only in the, probably rare, case when the relations between the cues and the criterion are not conditionally monotone, that the subjects may find that there is a better alternative to their linear model. Thus, if the cognitive system had only one model to use for judgment, a linear model is what it ought to use. Under these

circumstances, it is hardly surprising that this is indeed the most common model used by subjects. At the same time, using such a model is clearly very useful since it allows the subjects to make reasonably good predictions even if they have not learned from experience with the task.

Further studies showed, however, that functional rules, such as linear and U-shaped functions were by no means the only rules considered by subjects in these tasks, even though they tended to prefer these kinds of rules. Fig. 1 gives a summary of the results of our experiments.

As is shown in this figure, subjects prefer to assume that there is a rule, rather than that there is no rule, that this rule is deterministic, rather than probabilistic, that the values to be predicted from the cue values do in fact depend on these cue values, rather than on other aspects such as trial number, that the rule is functional, rather than nonfunctional, and that the rule is a positive linear function, rather than any other function. If the hypothesis about linearity fails, the subjects try some of the other functional rules before backtracking their decision tree, trying a nonfunctional rule, or a rule that does not assume cue dependence, such as a sequential rule, meaning that the ordinal number of the cue values, rather than their actual values, are used to form their hypothesis. For example, the subjects may believe that low criterion values are regularly followed by high criterion values, or *vice versa*. When these rules also fail, the subjects tend to assume that there is no rule at all, rather than to seriously consider the possibility that the rule may be probabilistic in character. Therefore, they go to a

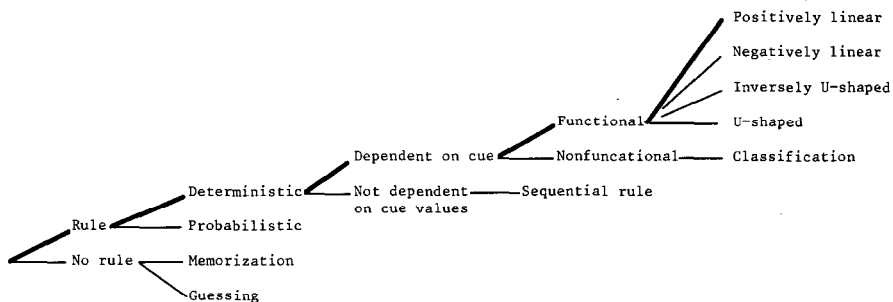


Fig. 1. Summary of results from studies on probabilistic inference tasks. Note that there is no further breakdown of probabilistic rules. This is because no such rules are observed in the experiments. The heavy line in the figure indicates the most common path.

memorization strategy, or they may just give up and guess.

The result that the subjects do not seem to consider the task to be probabilistic, or to use a strategy appropriate to a probabilistic task, agrees with that obtained under other kinds of circumstances by Tversky and Kahneman (1978). In this and other papers, they have shown that when subjects have to estimate probabilities, they generally do not follow the rules of probability calculus. Instead, they use various heuristics such as representativeness and availability. The results in Tversky and Kahneman (1978) strongly suggest that the subjects believe that the relations in the experiment are causal, rather than statistical in nature. Consequently, they use inference rules appropriate to deterministic tasks, rather than the rule appropriate to probabilistic tasks. In this context, it is interesting to note that Wason and Johnson-Laird (1972) earlier invoked the same kind of hypothesis to explain why subjects failed to follow adequate logical rules in another kind of task also requiring the use of a formal schema for making inferences. Specifically, Wason and Johnson-Laird found that subject did not understand the logical relation of implication ('if, then') and that they tended to consider this relation equivalent to a double implication ('if, and only if'), as would be appropriate for a causal task. They also showed that task content had a very strong effect on the subjects' inference rules, a finding consistent with those of Tversky and Kahneman (1978), but which has only rather limited support in probabilistic inference tasks requiring the subjects to use cues for making inferences (Brehmer and Kuylensstierna 1980). Taken together, however, these results provide considerable evidence that determinism, or causality, is a very basic schema used by people to make sense of the world.

While this may be a useful schema in many circumstances, it nevertheless leads to rather dramatic errors in many circumstances (see Tversky and Kahneman 1978), so it is clearly not universally applicable.

This leads to the important question of whether it is possible to improve the subjects' performance by informing them of the basic probabilistic character of the task. In a series of experiments we have studied the effects of giving this kind of information in inference tasks requiring the subjects to use probabilistic cues to make inferences about a criterion variable (Brehmer and Kuylensstierna 1978, 1980; Johansson and Brehmer 1979). Specifically, we have used various forms of instructions to explain to the subjects that the task is probabilistic, ranging from a simple statement that it is impossible to make perfect inferences

on every trial because the task is probabilistic to explicit graphical explanation in terms of scatter diagrams. This has not led to any improvement in performance or to any change in strategy. We have then tried to explain to the subjects, not only that the task is probabilistic, but also what kind of strategy is applicable to these kinds of tasks, *i.e.*, that they should look for the regression function and use it consistently. When this also failed, we thought that this may be because the subjects could not test their hypotheses about the rule for the task, so we therefore also informed the subjects about the magnitude of the errors to expect, given that they had found the optimal rule to help them evaluate their performance. Again the results were negative, as they were also when the tasks were given a content which clearly suggested that the relations between cues and criterion were probabilistic, and which was interpreted by the subjects to mean that the task was probabilistic, as assessed in a separate experiment. Regardless of instruction or task content, the subjects showed the typical behavior for these kinds of inference tasks: suboptimal performance due to inconsistent inference strategies with inconsistency being a monotone function of the uncertainty of the task.

These results, taken as a whole, strongly suggest that people simply do not have the cognitive schemata needed for efficient performance in probabilistic tasks. These experiments are, of course, laboratory experiments using college students as subjects. However, there is evidence that these results are not limited to the laboratory. Experienced clinicians behave in the same manner as the subjects in these experiments when making inferences in tasks with which they have years of experience (Brehmer 1976b).

One possible explanation why people do not learn in these tasks is that they are unable to test their hypotheses when the information comes in a nonsystematic way. However, this is, at best, only part of the explanation. There are results (Slovic personal communication) indicating that subjects do not learn probabilistic inference tasks more efficiently when the information is presented in a systematic way calculated to give them maximum information or when they are given an opportunity to select instances in whatever order they wish.

These results, then, support the earlier results on clinical inference is that they show that people do not learn optimal strategies from experience even if they are given massive amounts of practice. The reason why the subjects fail to improve in these tasks seems to be that they

lack the necessary basic schemata to help them understand and use the information provided by their experience. Rather than using the appropriate statistical schemata, subjects use an inappropriate causal or deterministic schema. Even though this does not, of course, prevent them from learning anything at all, the deterministic schema may be a good approximation to what is to be learned, it will certainly prevent them from learning as much as they could. However, even when subjects are using the appropriate statistical schemata, their performance is not better than when they are not using these schemata (Brehmer and Kuylenstierna 1979, 1980) perhaps because the subjects are not used to this way of dealing with inference tasks.

These findings raise the problem of why the subjects fail to detect the probabilistic nature of these tasks, which is so obvious to the experimenter.

Upon reflection this is, perhaps, not so strange. The characteristic of probabilism is, of course, not manifest, but it has to be inferred. This means, first, that the subject has to have an adequate notion of probabilism among this schemata for organizing his experience, and, second, that he has some criterion for when a task should be considered probabilistic. There is, of course, no absolute criterion of probabilism, so it can only be evaluated in comparison with some deterministic hypothesis. One way of doing this is to assume a functional, or other, rule and evaluate the fit of this rule. However, it is clear that this will put the subject in quite a bind, for to evaluate his rule, he has to make some assumption about the error in this system, and to evaluate the error, he has to make assumptions about the rule. It would seem, therefore, that the only way in which a subject could do this would be to try all his rules, and then shift to a statistical strategy, when he can find no more rules to try. Although the subjects certainly seem to follow the first part of this strategy, they fail to switch to the statistical rule when they have run out of deterministic rules. Instead the subjects seem to think that there is some deterministic rule for the task, but that they cannot find it. It is, of course, impossible to prove them wrong; the fact that they cannot find the rule does not prove that there is no rule. This illustrates that for a person with a firm belief in the deterministic character of the world, there is nothing in his experience that would force him to discover that the task is probabilistic and to give up the notion of determinism. To detect the probabilistic nature of the task, the subject clearly has to have the hypothesis that it is probabilistic.

If a subject cannot detect that a given task is probabilistic, it is not surprising that he does not have a general statistical approach. On the other hand, the notion of determinism is just as hidden from experience. It is therefore legitimate to ask why subjects do not have the opposite approach, *i.e.*, why they do not have probabilism as a basic schema and fail to detect when the task is deterministic. Such an approach ought to be at least as compatible with their experience as the deterministic approach.

Some observations by Piaget and Inhelder (1975) are relevant here. They point out that the notions of chance and probability cannot develop until a person has reached the stage of formal operations, and that it can develop only in contrast to the schema of order, or causality. They then document in a series of experiments that the notion of probability is indeed dependent on the nature of the person's general cognitive structures, and that it develops only after the schema of causality has been firmly established around the age of 11 to 12 years.

These results, then, suggest that the notion of order is developmentally prior to that of probabilism. If so, it is no longer surprising that the subject in our experiments do not entertain probabilistic hypotheses, especially in consideration of all the results which show that not very many people enter the stage of formal operations.

To summarize, our discussion shows that subjects do not seem to be able to learn to perform optimally in probabilistic inference tasks, and that this is due to lack of adequate schemata for handling the probabilistic aspect of the world. This may seem surprising, but it can be understood if we remember that the probabilism of task is not manifest, and that it can be detected only by a person who already has the notion of probabilism in his cognitive equipment. In short, probabilism must be invented before it can be detected. This, in turn, requires considerable cognitive development on the part of the person. We may get some appreciation for how hard it is to make this invention if we remember that probability calculus was not invented until the seventeenth century. One explanation why it had to take so long, consistent with the interpretation by Piaget and Inhelder of the development of the child, is that it was only at this point in time that the notion of causality had reached such a level that it could provide a suitable contrast against which to evaluate disorder. Be that as it may, the fact that probability calculus is such a late invention, despite that it would have been useful long before it was invented, is evidence that an adequate

understanding of probabilism is very hard to come by from our experience.

Learning when the outcomes are dependent upon judgment

So far, we have considered only the case when a person learns passively by observing the world. This, however, is not the only, or perhaps even the most common, case. Often, and this is certainly the case in clinical situations, a person has to act on his judgment, *i.e.*, by prescribing a cure, and the outcome from which he is to learn will depend not only on the quality of the person's judgment but also on the nature of his actions. The most fundamental characteristic of action is its selective nature; doing one thing precludes doing another. Thus, giving one kind of therapy precludes giving another. Furthermore, action may be selective also in that it is directed only at some of those who could have received it. For example, scarce resources may force us to treat only a limited number of all those who apply for treatment. As we shall see, the possibilities of learning from experience may be extremely limited in this case. Specifically, there will be a number of factors that prevent us from detecting that our judgment is fallible.

Consider the case where a psychologist has invented a new form of therapy. He now wishes to show its usefulness. However, he cannot give therapy to everyone who wants it, so he makes the reasonable decision to give therapy only to those who need it the most, *i.e.*, to those he judges to be most ill. He then goes on to give his therapy to those people and then makes a new judgment concerning the status of his patients. Almost inevitably, he will now find that they have improved, and he will be led to conclude that his treatment was useful, and that he made the correct judgment concerning his patients. The problem is that there is a very likely alternative explanation: that the improvement he observed is simply a regression effect. Since his judgment is likely to be unreliable (indeed, the results from studies on clinical inference would suggest that his judgment would most likely be grossly unreliable), it will change from the first to second time so that those patients given low judgments on the first occasion will, on the average, be given higher judgments on the second occasion, and *vice versa*. This is likely to be a common mechanism for it must often be the case that the resources are limited, forcing us to select cases for treatment, and if we select extreme cases, the regression effect will automatically insure that

our method will appear useful. Under these circumstances, nobody should be surprised that old methods of therapy are never given up but always found useful by clinical experience, and abandoned only for new methods presumably found useful for the very same reason as the old ones. Nor should anyone be surprised that as a treatment becomes more widely available, and thus is given to less severely disturbed patients, its effect will seem smaller.

Kahneman and Tversky (1973) have pointed to some tragical consequences of the regression effect: that it makes us use punishment rather than reward for influencing behavior. This is because punishment is likely to be given to those judged to be bad, *i.e.*, to those who appear to be low with respect to the characteristic being judged. These people are, of course, exactly those who would be judged to have improved from the first to the second occasion simply because of the regression effect. Rewards, on the other hand, are likely to be given to those who are judged to be good, *i.e.*, to those judged to have a high value. For these people, the regression effect will work in the opposite direction, so that they will seem to have a lower score on the second occasion. Thus, it is clear that one's experience will tend to prove that punishment is an effective means for improving performance, and that reward is not only ineffective, but perhaps even harmful. Kahneman and Tversky document by examples that this is indeed what people learn from their experience, contrary to what a person who has studied the psychological literature on the effects of reward and punishment would expect. A person's experience, it seems, will not necessarily tell the truth.

Let us now consider the opposite case, that our judgment has little or no validity, but we have a powerful and efficient treatment. This case has recently been analyzed in detail by Einhorn and Hogarth (1978). In their interesting and important paper, they show how, under these circumstances, we may easily be led to overestimate the validity of our judgment.

In part, the problems with this case should be well known to social scientists: it is simply the case of self-fulfilling prophesies. Einhorn (1979) uses the example of the effects of deciding to fund a grant application. Regardless of the validity of the judgment of value or deservedness of the research application, it is clear that those who get grants are going to produce more and probably also better research than those who do not get grants, thus confirming the judgment of the granting

agency that those who got the grants were indeed those who should have had them. There is, of course, no ground for saying anything about the validity of the judgment of the basis of these outcomes. To do so, it is necessary to disentangle the effects of the treatment, *i.e.*, giving the grant, from the judgment. But this can be done only by *not* acting on one's judgment, and by using some other means of resource allocation, *e.g.* random distribution. But this is, of course, exactly what is not likely to happen, especially when the resources are scarce. In such a situation we are thus not likely to find the true validity of our judgment.

As we have already pointed out, the problem of learning from our experience when we also have to act upon our judgment is that the very fact that we do something will preclude getting the information we need to evaluate our judgment. Action is always selective, meaning that as we select certain cases for treatment, we by that very act also select other cases who do not get treatment. The only cases we can observe, therefore, are the true positives and the false positives.

In many situations, this may not be such a problem. Given that the selection ratio is extreme, it is clear that we may actually get only true positives, even if the validity of our judgment is extremely low. Scarce resources thus tend to give us an inflated sense of judgment ability. As the resources get less scarce, and the selection ratio less extreme, our judgment is going to appear less and less good. Perhaps this is the reason why we now have so much more concern about unnecessary surgery than we had 20 years ago when the resources for hospital care were less adequate than they are today.

While the need to act may logically preclude learning about the validity of one's judgment, the psychological effects may not be as great as one might expect. The problem of learning about the validity of one's judgment in the present case is basically that of learning about the relation between two dichotomous variables. The judgment dichotomizes the independent variable into two categories: those who do and those who do not get treatment, and the dependent variable is dichotomized into those who succeed and those who do not, *e.g.*, those who get well and those who do not.

This particular learning task has received considerable attention (Jenkins and Ward 1965; Smedslund 1963; Ward and Jenkins 1965). The results show that the subjects, when learning these tasks, tend to focus only on the number of true positives, *i.e.*, they follow the same

strategy of using only confirming evidence as we have observed earlier. This is, of course, not very satisfactory from a logical point of view. It makes sense, however, when we consider that under natural circumstances, *e.g.*, when people have to learn about the validity of their judgment, they will not have access to all four outcomes. Thus, it may not be so surprising that people have not learned the optimal way of coping with tasks of this sort.

So far, we have mentioned two aspects of the task that may give us a false sense of judgmental ability: the efficiency of our treatments and the selection ratio. However, it should also be clear that the base rate of the trait being judged is extreme, this will also affect our sense of judgmental ability. For example, if 90% of the patients who come to a clinical hospital are schizophrenic, it is not a great achievement to be able to correctly identify 80% of all patients as schizophrenic although an 80% hit rate may sound impressive when we do not consider the base rate, and may thus lead us to trust our judgment.

It should also be clear that there will be interactions among judgmental validity, selection ratio, base rate, and the efficiency of our treatment, and that the rate of positive hits will vary with all four of these factors. Einhorn and Hogarth (1978) present a formal model of how these factors interact. The importance of this model for future research on judgment and decision-making can hardly be overestimated. It provides a needed first step towards a more adequate conception of what a person's experience really is like, and this is, of course, a prerequisite for a more adequate conception of what a person may learn from that experience. Hopefully, the work of Einhorn and Hogarth will inspire others to continue their work and provide us with a more complete picture of the nature of experience.

Conclusion

We started this paper by discussing some results that seemed to contradict one of our deepest convictions: that experience leads to better judgment and decisions. Our further analysis of the problem has shown, however, that our faith in experience is, if not totally without foundation, so at least far from well grounded. This is because it stems from an untenable conception of the nature of experience, a conception that assumes that truth is manifest and does not have to be

inferred. A more adequate understanding of the nature of experience leads to a more pessimistic view of what its effects may be. This, in turn, leads to less surprise about the results of studies on judgment and decision making. It seems that these results are exactly what they should be, and if we do not learn from experience, this is largely because experience often gives us very little information to learn from.

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